

Linear Regression with Multiple Regressors

SW Chapter 6

ECON3500: Econometrics and Applications

Spring 2026

Roadmap

Roadmap

Why Multiple Regression?

- Motivation and limitations of simple regression

The Multiple Regression Model

- Definition, OLS estimation, matrix form

Interpreting Coefficients

- Ceteris paribus interpretation
- Examples and knowledge checks

Omitted Variable Bias

- Derivation, formula, examples
- General cases

Measures of Fit

- R^2 and Adjusted R^2
- Model comparison

Least Squares Assumptions

Learning Objectives

! Central Question

How do we estimate the effect of X_1 on Y while **holding other factors constant**?

By the end of this chapter, you will:

Estimate and interpret **multiple regression** models

Understand **omitted variable bias** deeply

Calculate and interpret a new measure of fit, the **adjusted R^2**

Recognize and handle **multicollinearity**

Update our knowledge of the LS assumptions and sampling distributions of the OLS estimators when we have multiple regressors.

Why Multiple Regression?

The Limitation of Simple Regression

Recall our wage equation from Chapter 4:

$$wage_i = \beta_0 + \beta_1 \cdot education_i + u_i$$

Problem: What's in u_i ?

Work experience

Ability

Location

Industry

Gender, race, age...

If any of these are **correlated with education**, then $\hat{\beta}_1$ suffers from **omitted variable bias**!

Three Reasons for Multiple Regression

Control for confounders

- Reduce omitted variable bias
- Get closer to causal effects

More flexible functional forms

- Include X^2 and log-transformed variables for nonlinear relationships
- Interaction terms: $X_1 \cdot X_2$

Better predictions

- More information $\Rightarrow$ better forecasts

Example: Wage Equation

$$wage_i = \beta_0 + \beta_1 \cdot education_i + \beta_2 \cdot experience_i + u_i$$

β_1 : the effect of education on wage, holding experience constant

β_2 : the effect of experience on wage, holding education constant

u_i : error term - what we can't explain with education and experience

The Multiple Regression Model

Definition

i Multiple Linear Regression Model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

where:

Y_i : dependent variable (outcome)

$X_{1i}, \dots, X_{ki}$: k independent variables (regressors)

$\beta_0, \dots, \beta_k$: $k + 1$ unknown parameters

u_i : error term

OLS Estimation

Same principle as simple regression: Minimize sum of squared residuals

$$\min_{\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k} \sum_{i=1}^n u_i^2$$

where

$$u_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_{1i} - \dots - \hat{\beta}_k X_{ki}$$

In practice: We use statistical software (Stata, R, Python) to compute $\hat{\beta}_0, \dots, \hat{\beta}_k$

Interpreting Coefficients

The Ceteris Paribus Interpretation

! Partial Effect

β_j is the **partial effect** of X_j on Y , holding all other regressors constant.

Mathematically:

$$\beta_j = \frac{\partial Y}{\partial X_j} = \frac{\Delta Y}{\Delta X_j} \Big|_{\text{other } X\text{s fixed}}$$

In words:

“Holding $X_2, \dots, X_k$ constant...”

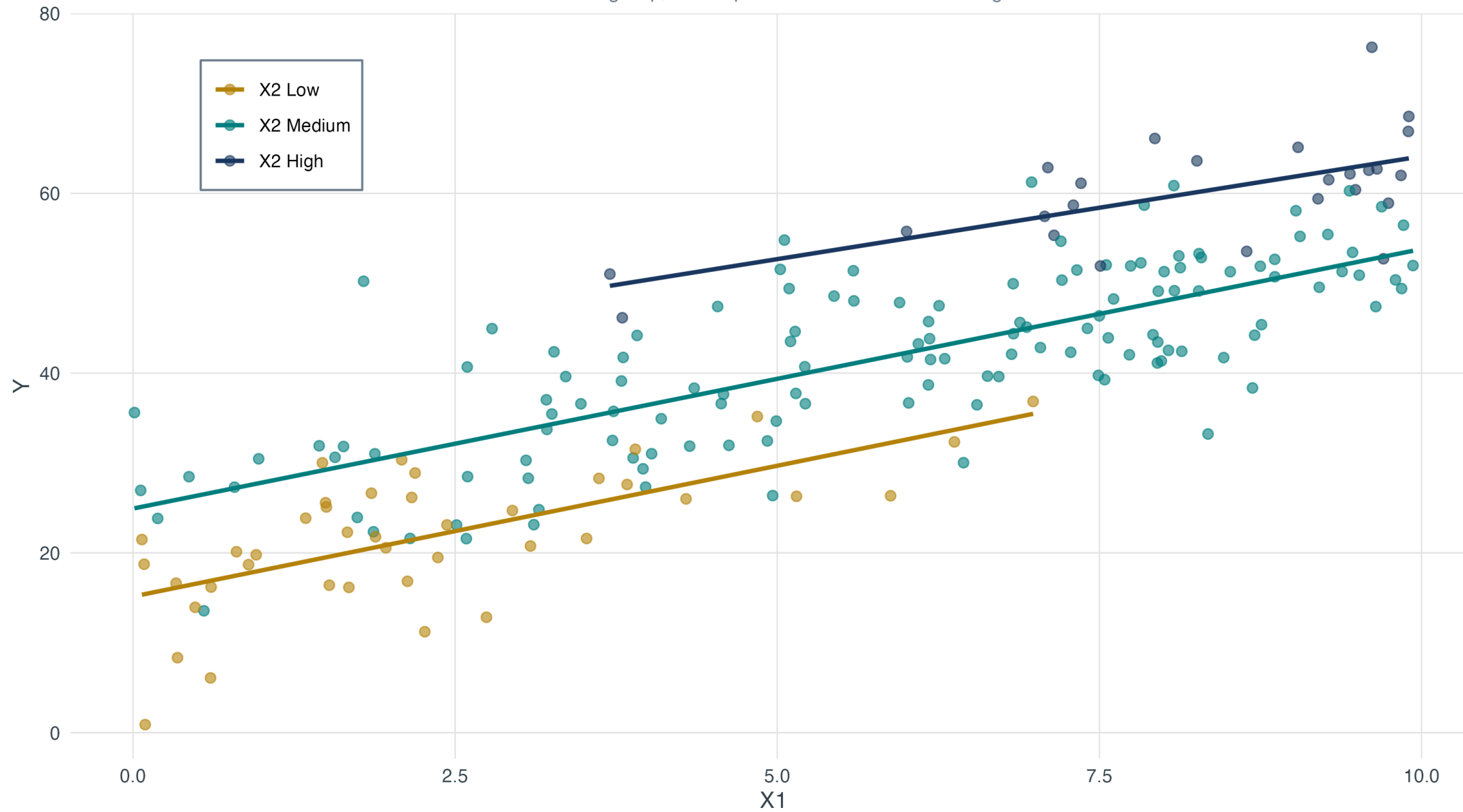
“Controlling for $X_2, \dots, X_k$...”

“Ceteris paribus” (all else equal)

Visualizing Ceteris Paribus

Ceteris Paribus: Holding Other Factors Constant

Within each X_2 group, the slope shows effect of X_1 holding X_2 fixed



Within each X_2 group, the slope shows the effect of X_1 holding X_2 fixed.

Example: Wage Equation

Estimated model:

$$\text{wage} = 5.2 + 2.1 \cdot \text{education} + 0.6 \cdot \text{experience}$$

Interpretation of $\hat{\beta}_1 = 2.1$:

Holding experience constant...

...each additional year of education is associated with...

...\$2.10 higher hourly wage

Interpretation of $\hat{\beta}_2 = 0.6$:

Holding education constant...

...each additional year of experience is associated with...

...\$0.60 higher hourly wage

Knowledge Check: Interpretation

Question

You estimate: $colGPA = 1.3 + 0.45 \cdot hsGPA + 0.009 \cdot ACT$

Compare two students:

Student A: $hsGPA = 3.5$, $ACT = 25$

Student B: $hsGPA = 4.0$, $ACT = 25$ (same ACT!)

What is the predicted difference in their college GPAs?

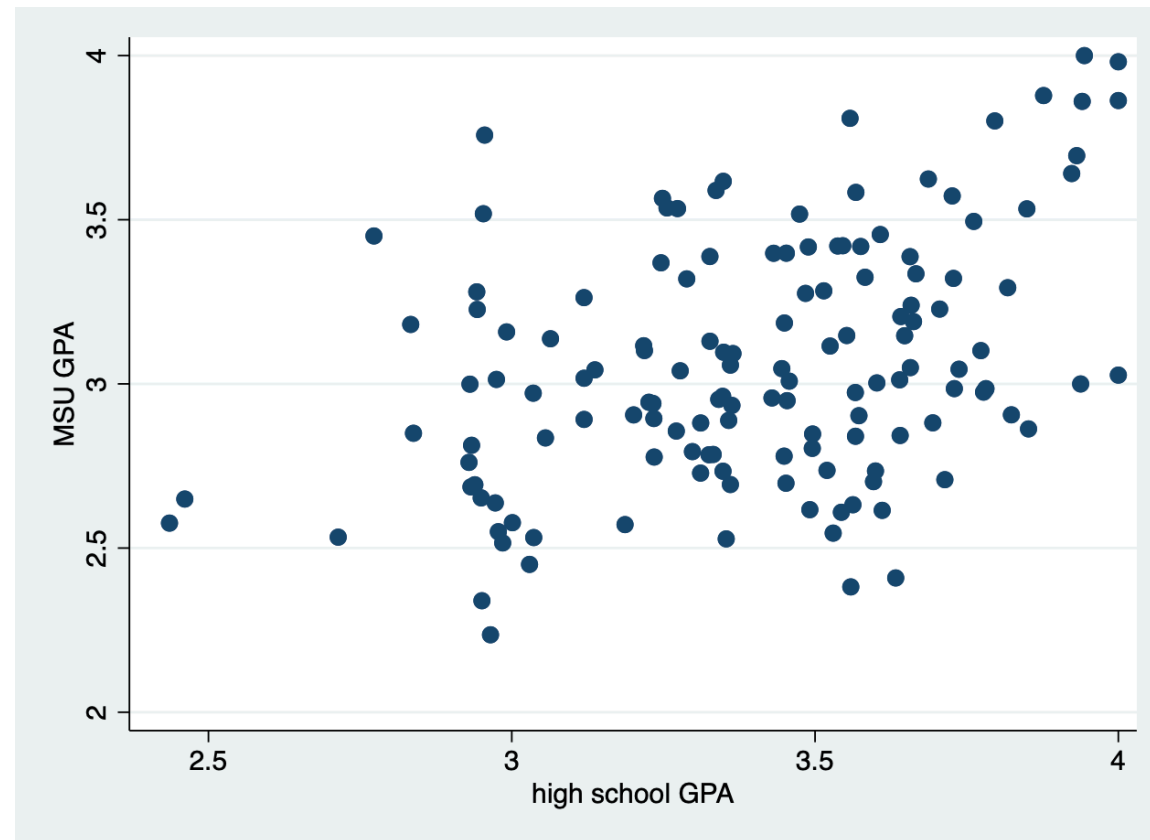
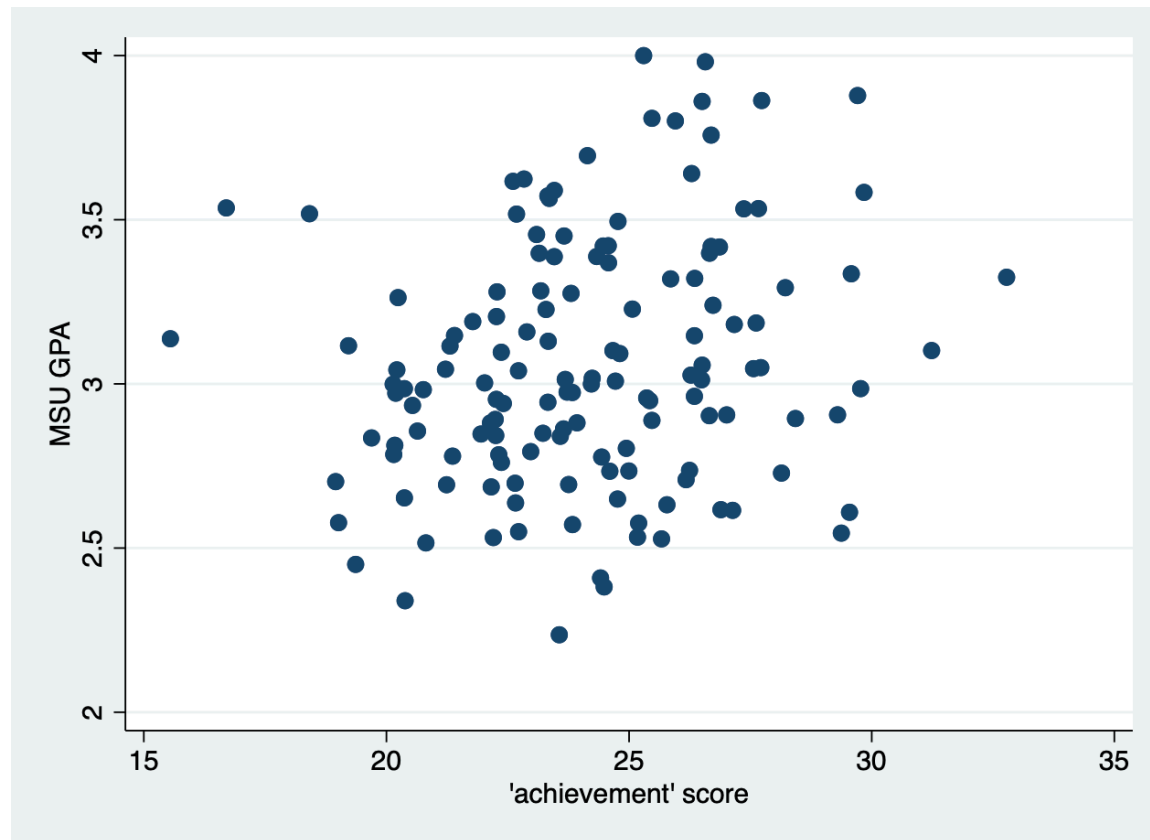
Answer

Difference = $0.45 \times (4.0 - 3.5) = 0.45 \times 0.5 = 0.225$

Student B is predicted to have a college GPA 0.225 points higher, **holding ACT constant**.

Example: Determinants of College GPA

Let's look at the relationship between high school and college GPA, controlling for test scores (*MSU students in Fall 1994*).



What predicts college GPA?

Example: Determinants of College GPA

We can set up the following **population multiple regression model**

$$colGPA_i = \beta_0 + \beta_1 hsGPA_i + \beta_2 ACT_i + u_i$$

Example: Determinants of College GPA

```
. regress colGPA hsGPA ACT
```

Source	SS	df	MS	Number of obs	=	141
Model	3.42365506	2	1.71182753	F(2, 138)	=	14.78
Residual	15.9824444	138	.115814814	Prob > F	=	0.0000
Total	19.4060994	140	.138614996	R-squared	=	0.1764
				Adj R-squared	=	0.1645
				Root MSE	=	.34032

colGPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
hsGPA	.4534559	.0958129	4.73	0.000	.2640047	.6429071
ACT	.009426	.0107772	0.87	0.383	-.0118838	.0307358
_cons	1.286328	.3408221	3.77	0.000	.612419	1.960237

Stata regression output for college GPA

Example: Determinants of College GPA

Estimated equation (or **OLS regression line**):

$$\widehat{colGPA} = 1.29 + 0.453 \cdot hsGPA + 0.0094 \cdot ACT$$

Interpretation:

Holding ACT fixed, an additional point in high school GPA is associated with 0.453 points higher in college GPA

Or: If we compare two students with the same ACT, but the *hsGPA* of student A is one point higher, we predict student A to have a *colGPA* that is 0.453 points higher than that of student B

Omitted Variable Bias

Omitting relevant variables: the simple case

Let's work through some theory! **True population model:**

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

Both X_1 and X_2 belong in the model.

But we estimate (omitting X_2):

$$Y_i = \beta_0^{\sim} + \beta_1^{\sim} X_{1i} + u_i^{\sim}$$

! Question

Is $\beta_1^{\sim}$ an unbiased estimator of β_1 ?

Deriving the Bias

Assume X_2 is linearly related to X_1 :

$$X_{2i} = \delta_0 + \delta_1 X_{1i} + v_i$$

Substitute into the true model:

Step 1: Start with the true model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i$$

Step 2: Replace X_{2i} with its relationship to X_1 :

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 (\delta_0 + \delta_1 X_{1i} + v_i) + u_i$$

Step 3: Distribute β_2 :

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 \delta_0 + \beta_2 \delta_1 X_{1i} + \beta_2 v_i + u_i$$

Step 4: Collect like terms:

$$Y_i = (\beta_0 + \beta_2 \delta_0) + (\beta_1 + \beta_2 \delta_1)X_{1i} + (\beta_2 v_i + u_i)$$

! Omitted Variable Bias Formula

$$E[\tilde{\beta}_1] = \beta_1 + \beta_2 \delta_1$$

where $\beta_2 \delta_1$ is the **bias**.

When Does OVB Occur?

$$\text{Bias} = \beta_2 \delta_1$$

Bias is ZERO if:

$\beta_2 = 0$ (omitted variable is irrelevant)

$\delta_1 = 0$ (omitted variable uncorrelated with X_1)

Bias is NON-ZERO when:

Omitted variable belongs in the model ($\beta_2 \neq 0$), AND

Omitted variable is correlated with included variable ($\delta_1 \neq 0$)

Signing the Direction of Bias

Signing the Direction of Omitted Variable Bias

Bias = $\beta_2 \times \delta_1$, where δ_1 has sign of $\text{Corr}(X_1, X_2)$

$\text{Corr}(X_1, X_2) > 0$

$\text{Corr}(X_1, X_2) < 0$

$\beta_2 > 0$

**Positive
Bias ↑**

**Negative
Bias ↓**

$\beta_2 < 0$

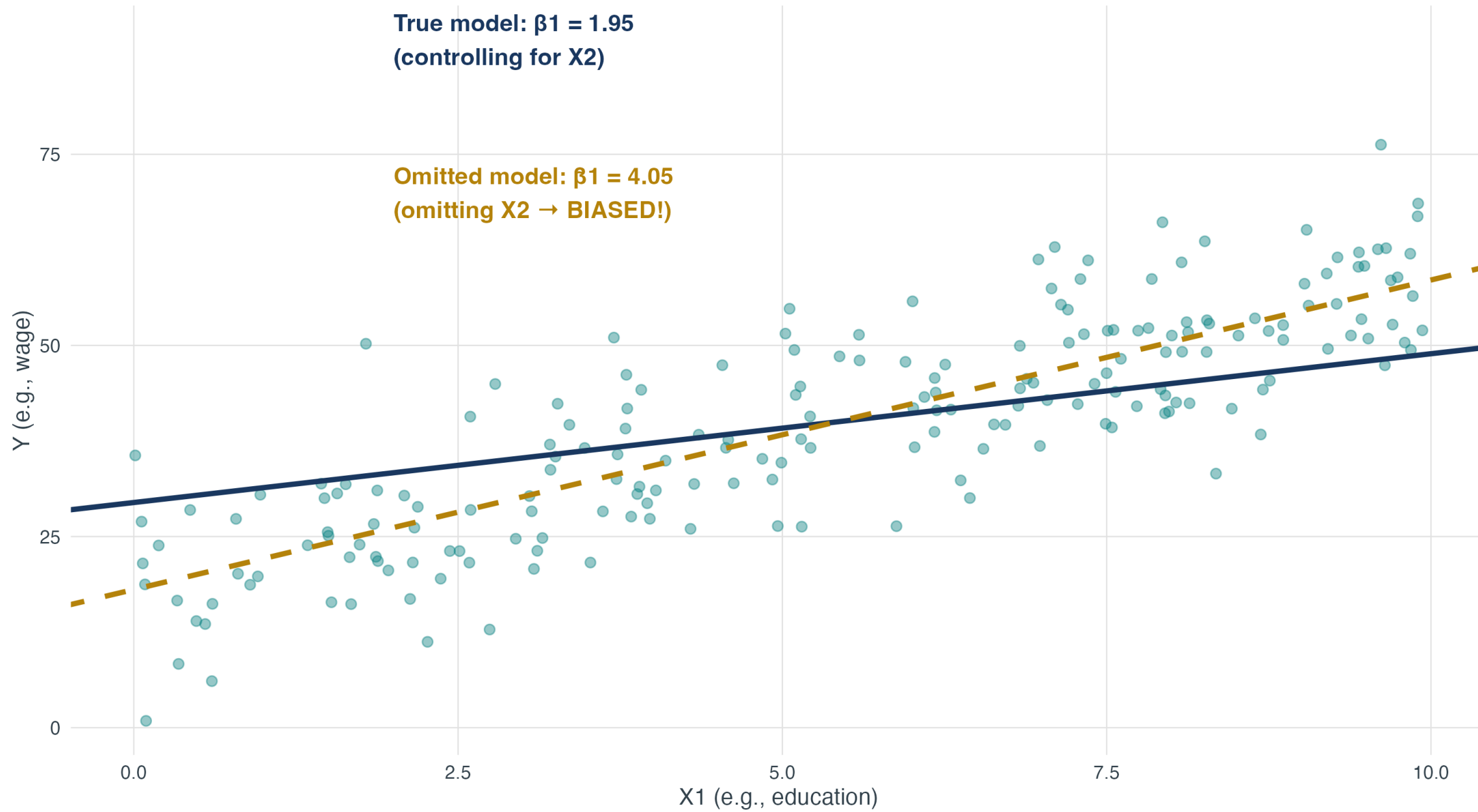
**Negative
Bias ↓**

**Positive
Bias ↑**

Visualizing Omitted Variable Bias

Omitted Variable Bias

Bias = 2.10 (overstates true effect)



Example: Returns to Education

True model:

$$wage = \beta_0 + \beta_1 \cdot educ + \beta_2 \cdot abil + u$$

But ability is unobserved!

Relationship between ability and education:

$$abil = \delta_0 + \delta_1 \cdot educ + v$$

Substitute into the true model:

$$\begin{aligned} wage &= \beta_0 + \beta_1 \cdot educ + \beta_2(\delta_0 + \delta_1 \cdot educ + v) + u \\ &= (\beta_0 + \beta_2\delta_0) + (\beta_1 + \beta_2\delta_1) \cdot educ + (\beta_2v + u) \end{aligned}$$

What's the bias?

$\beta_2 > 0$ (higher ability $\Rightarrow$ higher wage)

$\delta_1 > 0$ (smarter people get more education) $\Rightarrow \beta_2 \delta_1 > 0$

The return to education β_1 will be **overestimated** because $\beta_2 \delta_1 > 0$

It will look as if people with many years of education earn very high wages, but this is partly due to the fact that people with more education are also more able on average.

Knowledge Check: OVB Direction

Question

You're studying the effect of police officers (X_1) on crime (Y).

True model: $crime = \beta_0 + \beta_1 \cdot police + \beta_2 \cdot poverty + u$

You omit poverty. What's the likely direction of bias on $\hat{\beta}_1$?

Hints:

Does poverty increase crime? (sign of β_2 ?) {.fragment}

Do high-poverty areas have more police? (sign of correlation?) {.fragment}

Answer

$\beta_2 > 0$ (poverty increases crime) {.fragment}

$\text{Corr}(police, poverty) > 0$ (more police where poverty is high) {.fragment}

Bias = $\beta_2 \times \delta_1 > 0 \Rightarrow$ **Positive bias** {.fragment}

Omitting poverty makes it look like police *cause* crime (reverse causality illusion)! {.fragment}

OVB: More General Cases

We can extend this intuition when we add more independent variables:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

But we estimate:

$$\tilde{y} = \tilde{\beta}_0 + \tilde{\beta}_1 x_1 + \tilde{\beta}_2 x_2$$

Key points:

No general statements possible about direction of bias

Can assume one regressor uncorrelated with others to make analysis tractable

Measures of Fit

R^2 in Multiple Regression

Same definition as simple regression:

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS}$$

Problem: R^2 **never decreases** when you add regressors!

Even if the new variable is totally irrelevant

Even if it's just random noise

R^2 mechanically increases (or stays constant)

This makes R^2 useless for comparing models with different numbers of regressors.

Adjusted R^2

i Adjusted R^2

$$\bar{R}^2 = 1 - \frac{n-1}{n-k-1} \cdot \frac{SSR}{TSS}$$

where k is the number of regressors (not including intercept).

Key property: $\bar{R}^2$ **penalizes** you for adding regressors

If new variable is relevant: $\bar{R}^2$ increases

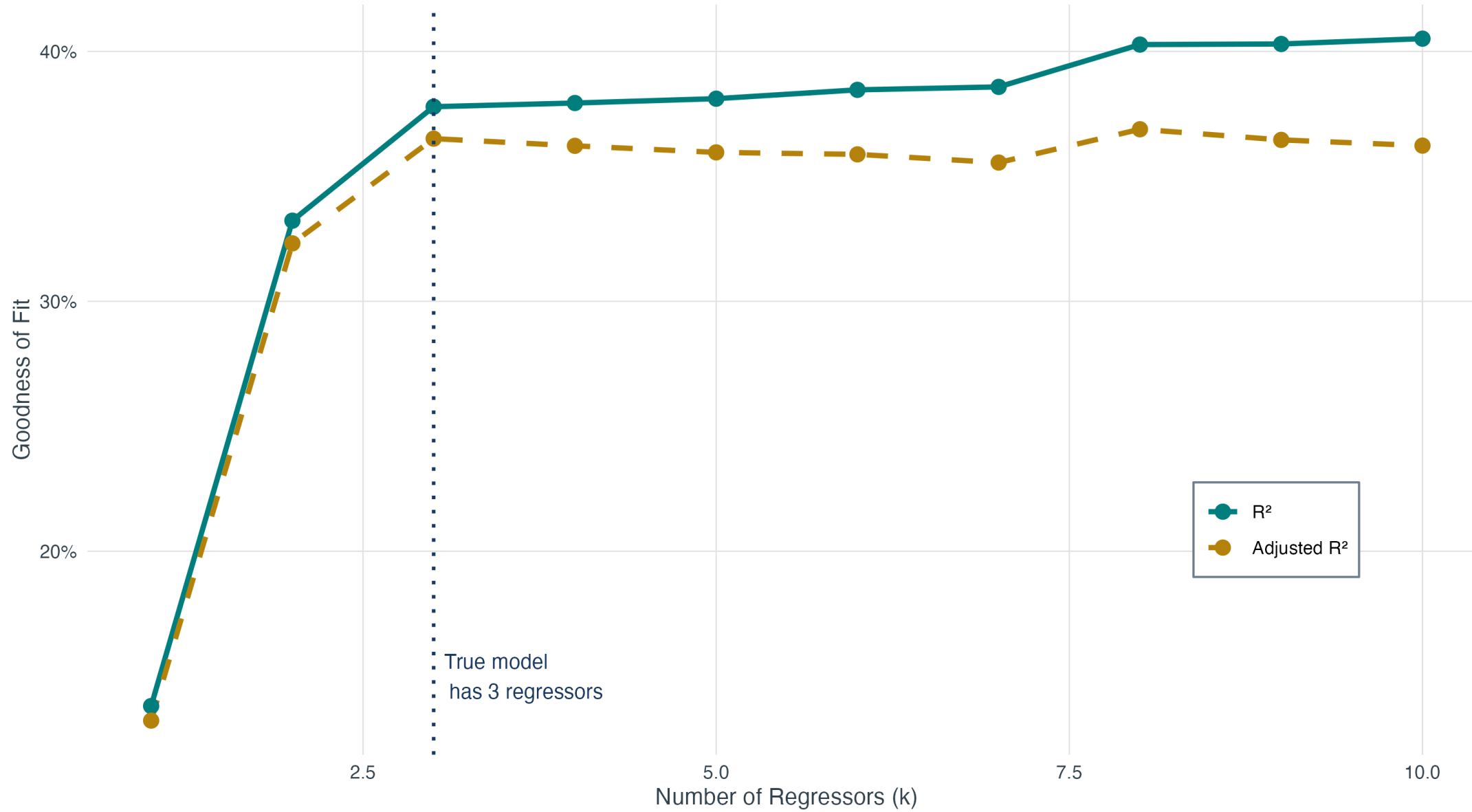
If new variable is irrelevant: $\bar{R}^2$ decreases (or barely changes)

$\bar{R}^2 < R^2$ always

R^2 vs Adjusted R^2

R^2 Always Increases, Adjusted R^2 Can Decrease

Adding irrelevant variables inflates R^2 but decreases adjusted R^2



R^2 always increases with k ; adjusted R^2 can decrease

Knowledge Check: Calculating Adjusted R^2

Question

You estimate a model with $n = 100$ observations and $k = 3$ regressors.

You obtain: $SSR = 200$ and $TSS = 500$

Calculate the adjusted R^2 .

Answer

$$\bar{R}^2 = 0.5875$$

Note: $R^2 = 0.6$, so $\bar{R}^2 < R^2$ as expected.

Knowledge Check: Calculating Adjusted R² - Step by Step

Given: $n = 100$, $k = 3$, $SSR = 200$, $TSS = 500$

Step 1: Write the formula:

$$\bar{R}^2 = 1 - \frac{n - 1}{n - k - 1} \cdot \frac{SSR}{TSS}$$

Step 2: Plug in the values:

$$\bar{R}^2 = 1 - \frac{100 - 1}{100 - 3 - 1} \cdot \frac{200}{500}$$

Step 3: Simplify:

$$\bar{R}^2 = 1 - \frac{99}{96} \cdot \frac{200}{500}$$

Step 4: Calculate:

$$\begin{aligned}\bar{R}^2 &= 1 - 1.03125 \times 0.4 \\ &= 1 - 0.4125 = 0.5875\end{aligned}$$

Check: $R^2 = 1 - \frac{200}{500} = 0.6$, so $\bar{R}^2 < R^2$ as expected.

Using Adjusted R^2 in Practice

Model 1:

```
regress wage education experience  
 $R^2 = 0.35$ , Adjusted  $R^2 = 0.34$ 
```

Model 2 (adding zodiac sign):

```
regress wage education experience zodiac_sign  
 $R^2 = 0.351$ , Adjusted  $R^2 = 0.339$ 
```

Conclusion:

R^2 increased slightly (always does)

Adjusted R^2 **decreased** $\Rightarrow$ zodiac sign is not useful!

Least Squares Assumptions

The Four LS Assumptions for Multiple Regression

We add one more assumption as we upgrade to the multiple regression model

Zero conditional mean: $E[u_i | X_{1i}, \dots, X_{ki}] = 0$

i.i.d. sampling: $(X_{1i}, \dots, X_{ki}, Y_i)$ are independently and identically distributed

Large outliers rare: X s and Y have finite fourth moments

No perfect multicollinearity: No X is a perfect linear combination of others

Assumption 1: Zero Conditional Mean (Extended)

$$E[u_i | X_{1i}, \dots, X_{ki}] = 0$$

Same interpretation as before, but now:

Error is uncorrelated with **all** regressors

Failure $\Rightarrow$ omitted variable bias

- If an omitted variable belongs in the equation (and is in u) and it is correlated with an included X , then this condition fails!

Best solution: include the omitted variable!

Assumptions 2 and 3

Assumption 2 (X s and Y are i.i.d.):

Satisfied automatically if the data are collected by simple random sampling

Assumption 3: Large outliers are rare:

Same as before: X s and Y have finite fourth moments

Check your data (scatterplots!) to make sure no crazy values

Assumption 4: No Perfect Multicollinearity (New!)

Perfect Multicollinearity

One regressor is an **exact** linear function of one or more other regressors.

Examples:

$$F = \frac{9}{5}C + 32$$

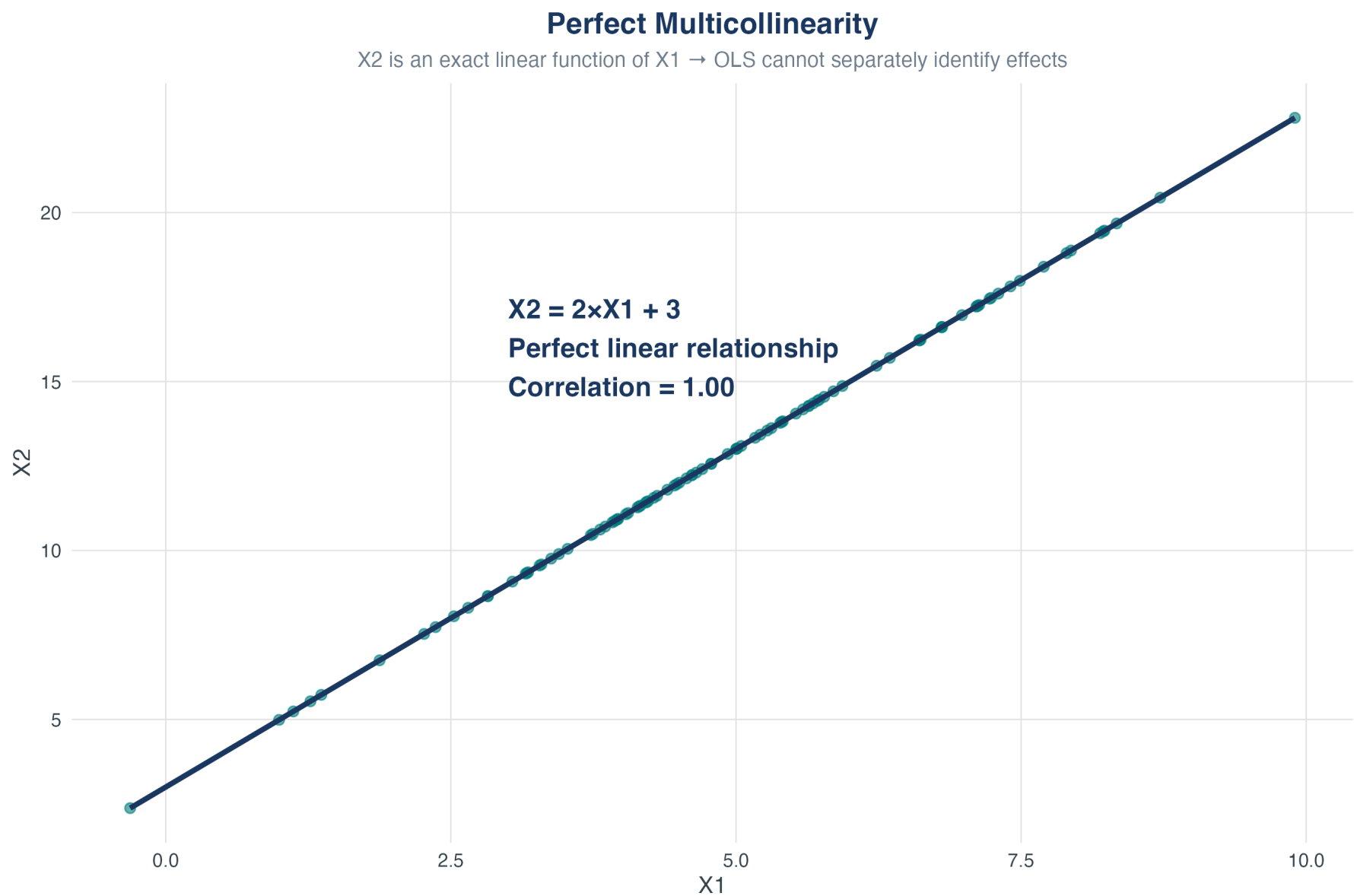
Including height in inches AND height in centimeters

Dummy variable trap: including all categories + intercept

What happens? OLS cannot be computed (infinite solutions)

What Stata does: Automatically drops one of the collinear variables (but which one will it be?)

Perfect Multicollinearity



Perfect Multicollinearity

```
.      gen ACT_36 = ACT/36
```

```
.  regress colGPA hsGPA ACT ACT_36
```

```
note: ACT_36 omitted because of collinearity
```

Source	SS	df	MS	Number of obs	=	141
				F(2, 138)	=	14.78
Model	3.42365506	2	1.71182753	Prob > F	=	0.0000
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				Adj R-squared	=	0.1645
Total	19.4060994	140	.138614996	Root MSE	=	.34032

colGPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
hsGPA	.4534559	.0958129	4.73	0.000	.2640047	.6429071
ACT	.009426	.0107772	0.87	0.383	-.0118838	.0307358
ACT_36	0	(omitted)				
_cons	1.286328	.3408221	3.77	0.000	.612419	1.960237

Perfect Multicollinearity

```
. gen lowhsGPA = hsGPA < 2
```

```
. regress colGPA hsGPA lowhsGPA ACT
```

```
note: lowhsGPA omitted because of collinearity
```

Source	SS	df	MS	Number of obs	=	141
Model	3.42365506	2	1.71182753	F(2, 138)	=	14.78
Residual	15.9824444	138	.115814814	Prob > F	=	0.0000
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colGPA	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
hsGPA	.4534559	.0958129	4.73	0.000	.2640047	.6429071
lowhsGPA	0	(omitted)				
ACT	.009426	.0107772	0.87	0.383	-.0118838	.0307358
_cons	1.286328	.3408221	3.77	0.000	.612419	1.960237

Perfect Multicollinearity - Dummy Variable Trap

Here we have a **dummy variable trap**:

$$\begin{aligned} colGPA = & \beta_0 + \beta_1 fresh + \beta_2 soph + \beta_3 junior \\ & + \beta_4 senior + \beta_5 hsGPA + u \end{aligned}$$

Perfect Multicollinearity - Dummy Variable Trap

```
. regress colGPA fresh soph jun senior hsGPA,robust
note: fresh omitted because of collinearity
```

```
Linear regression               Number of obs   =       141
                               F(4, 136)       =        6.66
                               Prob > F         =       0.0001
                               R-squared        =       0.1734
                               Root MSE     =       .34344
```

colGPA	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
fresh	0	(omitted)				
soph	.0714571	.3010712	0.24	0.813	-.5239295	.6668436
junior	-.0086131	.0914072	-0.09	0.925	-.1893764	.1721503
senior	-.0224848	.0881555	-0.26	0.799	-.1968178	.1518482
hsGPA	.4739247	.1003441	4.72	0.000	.2754881	.6723613
_cons	1.457486	.3277041	4.45	0.000	.8094308	2.105541

Multicollinearity

Imperfect Multicollinearity

Imperfect Multicollinearity

Two or more regressors are **highly** (but not perfectly) correlated.

Example: Including education AND test scores

Correlation ≈ 0.85 (high, but not 1.0)

Both can stay in the model

But estimates will be imprecise

Why Is Imperfect Multicollinearity a Problem?

Intuition:

$\hat{\beta}_1$ estimates effect of X_1 **holding** X_2 **constant**

But if X_1 and X_2 are highly correlated...

...there's very little variation in X_1 once X_2 is held constant

Data don't have much information about what happens when X_1 changes but X_2 doesn't

Consequence: Large standard errors for $\hat{\beta}_1$ and $\hat{\beta}_2$

Coefficients not statistically significant

This is **correct** — we genuinely can't tell them apart with these data!

What to Do About Multicollinearity

! Key Point

Imperfect multicollinearity is **not a violation** of OLS assumptions. It's a **data problem**, not a method problem.

Solutions:

Get more data (increases precision)

Drop one of the highly correlated variables (loses information, but may be necessary)

Combine variables (e.g., create an index)

Accept large standard errors (honest reflection of uncertainty!)

What NOT to do: Ignore it or claim the model is “broken”

Implementing Multiple Regression in Stata

Basic Stata Syntax

Multiple regression:

```
regress Y X1 X2 X3, robust
```

Example:

```
regress wage education experience, robust
```

Output interpretation:

Each coefficient is a **partial effect**

Standard errors are robust to heteroskedasticity

R^2 and adjusted R^2 shown at bottom

Detecting Multicollinearity in Stata

Check correlations:

```
correlate X1 X2 X3
```

Bringing It All Together

The Multiple Regression Workflow

Specify model: Which variables belong?

Estimate: Run OLS with robust SEs

Interpret: Partial effects (ceteris paribus)

Assess fit: Use adjusted R^2 for comparisons

Check assumptions: Zero conditional mean, no perfect multicollinearity

Beware OVB: Did you omit important variables?

Common Pitfalls

Forgetting “holding other variables constant”

- Always interpret coefficients as partial effects

Using R^2 instead of adjusted R^2

- R^2 is misleading when comparing models

Ignoring omitted variable bias

- Ask: what's in u ? Is it correlated with X ?

Panicking about multicollinearity

- Large SEs are honest! They reflect genuine uncertainty

Gauss-Markov in Multiple Regression

In Chapter 5, we learned that OLS is BLUE (Best Linear Unbiased Estimator) under the Gauss-Markov conditions.

The same result extends to multiple regression — we just need **one more assumption**:

i The Five Conditions for BLUE (Multiple Regression)

- . **Zero conditional mean:** $E[u_i | X_{1i}, \dots, X_{ki}] = 0$
- . **i.i.d. sampling**
- . **Large outliers rare** (finite fourth moments)
- . **No perfect multicollinearity**
- . **Homoskedasticity:** $\text{Var}(u_i | X_{1i}, \dots, X_{ki}) = \sigma_u^2$

Assumptions 1–4 → OLS is **unbiased**

Add assumption 5 → OLS is **efficient** (smallest variance among linear unbiased estimators)

If homoskedasticity fails, OLS is still unbiased — but no longer “Best”

Key Formulas Summary

Concept	Formula
Multiple regression	$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k + u$
Partial effect	$\beta_j = \frac{\partial Y}{\partial X_j} \text{ (other } X\text{s fixed)}$
OVB formula	$E[\tilde{\beta}_1] = \beta_1 + \beta_2 \delta_1$
Adjusted R^2	$\bar{R}^2 = 1 - \frac{n-1}{n-k-1} \frac{SSR}{TSS}$